

Movie Dataset Visualization Assessment

Exercise 1

A movie production company wants to understand whether spending more money on a movie leads to higher financial success and better audience reception.

Tasks

1. Create a new feature called **ROI**:

$$\text{ROI} = (\text{gross} - \text{budget}) / \text{budget}$$

2. Remove movies with missing values in `budget`, `gross`, or `score`.
3. Create a scatter plot that simultaneously shows:
 - Budget
 - Gross Revenue
 - Movie Score
 - Number of Votes
4. Apply appropriate transformations if necessary to improve readability.
5. Identify and annotate the **10 movies with the highest ROI**.
6. Create a second visualization that compares ROI distributions across the **top 10 genres by movie count**.
7. Based on your visualizations, answer the following questions:
 - Do larger budgets generally result in higher revenue?
 - Which genre appears to generate the highest returns relative to investment?
 - Are highly rated movies always financially successful?

Exercise 2

A streaming platform wants to identify the most reliable directors and production companies for future investments.

Tasks

1. Consider only:
 - Directors with at least 5 movies.
 - Companies with at least 20 movies.
2. For each director, calculate:
 - Number of Movies
 - Average Score
 - Median Score
 - Standard Deviation of Score
3. Create a visualization that helps distinguish:
 - Consistently high-performing directors
 - Inconsistent directors
 - Highly productive directors
4. For production companies, calculate:
 - Average Gross Revenue
 - Average ROI
 - Average Score
 - Number of Movies Produced
5. Create a visualization that displays all four metrics simultaneously.
6. Using your visualizations:
 - Identify three directors who are consistently successful.
 - Identify companies that prioritize quality over profitability.
 - Identify companies that prioritize profitability over quality.

Exercise 3

A film analyst wants to discover how movie preferences and industry trends have changed over time across genres and countries.

Tasks

1. Create decade groups from the movie release year.
2. For each decade and genre, calculate:
 - Number of Movies
 - Average Score
 - Total Gross Revenue
3. Build a visualization that clearly shows how genre popularity evolved across decades.
4. Consider only countries with at least 30 movies.
5. For each country, calculate:
 - Average Score
 - Average Budget
 - Average Gross Revenue
 - Average ROI
6. Create a single visualization that compares countries using all four metrics.
7. Define a **Hidden Gem** as:

```
score >= 8  
votes < median(votes)
```

8. Calculate the number of hidden gems produced by each genre and country.
 9. Create a visualization that highlights where hidden gems are most commonly found.
 10. Based on your analysis, answer the following questions:
 - Which genres gained popularity over time?
 - Which countries produce the highest-quality movies relative to budget?
 - Which genre and country combinations appear to be the most underrated by audiences?
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Submission Requirements

- All visualizations must contain appropriate titles, axis labels, legends, and annotations where necessary.
- Use suitable visualization types for each task and justify your choices.
- Apply data cleaning and preprocessing before visualization.
- Provide written insights and interpretations for every visualization.
- Clearly state any assumptions made during analysis.
- Submit both the source code and a report summarizing findings.